TODAY'S AGENDA

Aggregations + Group By

String / Date / Time Operations

Output Control + Redirection

Window Functions

Nested Queries

Lateral Joins

Common Table Expressions

EXAMPLE DATABASE

student(sid,name,login,gpa)

sid	name	login	age	gpa
53666	RZA	rza@cs	55	4.0
53688	Taylor	swift@cs	27	3.9
53655	Tupac	shakur@cs	25	3.5

enrolled(sid,cid,grade)

sid	cid	grade
53666	15-445	C
53688	15-721	A
53688	15-826	B
53655	15-445	B
53666	15-721	C

course(cid,name)

cid	name
15-445	Database Systems
15-721	Advanced Database Systems
15-826	Data Mining
15-799	Special Topics in Databases

AGGREGATES

Functions that return a single value from a bag of tuples:

→ **AVG(col)** → Return the average col value.
→ **MIN(col)** → Return minimum col value.
→ **MAX(col)** → Return maximum col value.
→ **SUM(col)** → Return sum of values in col.
→ **COUNT(col)** → Return # of values for col.

AGGREGATES

Aggregate functions can (almost) only be used in the **SELECT** output list.

Get # of students with a "@cs" login:

```
SELECT COUNT(login) AS cnt
  FROM student WHERE login LIKE '%@cs'
```

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```
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```

```
SELECT COUNT(*) AS cnt
```

```
SELECT COUNT(1) AS cnt
  FROM student WHERE login LIKE '%@cs'
```

AGGREGATES

Aggregate functions can (almost) only be used in the **SELECT** output list.

Get # of students with a "@cs" login:

```
SELECT COUNT(login) AS cnt
```

```
SELECT COUNT(*) AS cnt
```

```
SELECT COUNT(1) AS cnt
```

```
SELECT COUNT(1+1+1) AS cnt
  FROM student WHERE login LIKE '%@cs'
```

MULTIPLE AGGREGATES

Get the number of students and their average GPA that have a "@cs" login.

```
SELECT AVG(gpa), COUNT(sid)
  FROM student WHERE login LIKE '%@cs'
```

AVG(gpa)	COUNT(sid)
3.8	3

AGGREGATES

Output of other columns outside of an aggregate is undefined.

Get the average GPA of students enrolled in each course.

```
SELECT AVG(s.gpa), e.cid
  FROM enrolled AS e JOIN student AS s
    ON e.sid = s.sid
```

AVG(s.gpa)	e.cid
3.86	???

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```
SELECT AVG(s.gpa), e.cid
  FROM enrolled AS e JOIN student AS s
    ON e.sid = s.sid
```

AVG(s.gpa)	e.cid
3.86	???

```
SELECT AVG(s.gpa), ANY_VALUE(e.cid)
  FROM enrolled AS e JOIN student AS s
    ON e.sid = s.sid
```

AVG(s.gpa)	e.cid
3.86	15-445

GROUP BY

Project tuples into subsets and calculate aggregates against each subset.

```
SELECT AVG(s.gpa), e.cid
  FROM enrolled AS e JOIN student AS s
    ON e.sid = s.sid
 GROUP BY e.cid
```

e.sid	s.sid	s.gpa	e.cid
53435	53435	2.25	15-721
53439	53439	2.70	15-721
56023	56023	2.75	15-826
59439	59439	3.90	15-826
53961	53961	3.50	15-826
58345	58345	1.89	15-445

AVG(s.gpa)	e.cid
2.46	15-721
3.39	15-826
1.89	15-445

GROUP BY

Non-aggregated values in **SELECT** output clause must appear in **GROUP BY** clause.

```
SELECT AVG(s.gpa), e.cid, s.name
  FROM enrolled AS e, student AS s
 WHERE e.sid = s.sid
 GROUP BY e.cid
```

GROUP BY

Non-aggregated values in **SELECT** output clause must appear in **GROUP BY** clause.

```
SELECT AVG(s.gpa), e.cid, s.name
  FROM enrolled AS e, student AS s
 WHERE e.sid = s.sid
 GROUP BY e.cid
```

GROUP BY

Non-aggregated values in **SELECT** output clause must appear in **GROUP BY** clause.

```
SELECT AVG(s.gpa), e.cid, s.name
  FROM enrolled AS e JOIN student AS s
    ON e.sid = s.sid
 GROUP BY e.cid, s.name
```

HAVING

Filters results based on aggregation computation.

Like a **WHERE** clause for a **GROUP BY**

```
SELECT AVG(s.gpa) AS avg_gpa, e.cid
  FROM enrolled AS e, student AS s
 WHERE e.sid = s.sid
   AND avg_gpa > 3.9
 GROUP BY e.cid
```

HAVING

Filters results based on aggregation computation.

Like a **WHERE** clause for a **GROUP BY**

```
SELECT AVG(s.gpa) AS avg_gpa, e.cid
  FROM enrolled AS e, student AS s
 WHERE e.sid = s.sid
 GROUP BY e.cid
 HAVING avg_gpa > 3.9;
```

HAVING

Filters results based on aggregation computation.

Like a **WHERE** clause for a **GROUP BY**

```
SELECT AVG(s.gpa) AS avg_gpa, e.cid
  FROM enrolled AS e, student AS s
 WHERE e.sid = s.sid
 GROUP BY e.cid
 HAVING AVG(s.gpa) > 3.9;
```

AVG(s.gpa)	e.cid
3.75	15-415
3.950000	15-721
3.900000	15-826

avg_gpa	e.cid
3.950000	15-721

STRING OPERATIONS

	String Case	String Quotes
SQL-92	**Sensitive**	**Single Only**
Postgres	Sensitive	Single Only
MySQL	Insensitive	Single/Double
SQLite	Sensitive	Single/Double
MSSQL	Sensitive	Single Only
Oracle	Sensitive	Single Only

```
WHERE UPPER(name) = UPPER('TuPaC')
```
SQL-92

```
WHERE name = "TuPaC"
```
MySQL

STRING OPERATIONS

LIKE is used for string matching.

String-matching operators

→ **'%'** Matches any substring (including empty strings).

→ **'_'** Match any one character

```
SELECT * FROM enrolled AS e
 WHERE e.cid LIKE '15-%'
```

```
SELECT * FROM student AS s
 WHERE s.login LIKE '%@c_'
```

STRING OPERATIONS

SQL-92 defines string functions.

→ Many DBMSs also have their own unique functions

Can be used in either output and predicates:

```
SELECT SUBSTRING(name,1,5) AS abbrv_name
  FROM student WHERE sid = 53688
```

```
SELECT * FROM student AS s
 WHERE UPPER(s.name) LIKE 'KAN%'
```

STRING OPERATIONS

SQL standard defines the || operator for concatenating two or more strings together.

SQL-92

```
SELECT name FROM student
 WHERE login = LOWER(name) || '@cs'
```

MSSQL

```
SELECT name FROM student
 WHERE login = LOWER(name) + '@cs'
```

MySQL

```
SELECT name FROM student
 WHERE login = CONCAT(LOWER(name), '@cs')
```

DATE/TIME OPERATIONS

Operations to manipulate and modify **DATE**/**TIME** attributes.

Can be used in both output and predicates.

Support/syntax varies wildly…

Demo: Get the # of days since the beginning of the year.

OUTPUT REDIRECTION

Store query results in another table:

→ Table must not already be defined.

→ Table will have the same # of columns with the same types as the input.

SQL-92

```
SELECT DISTINCT cid INTO CourseIds
  FROM enrolled;
```

Postgres

```
SELECT DISTINCT cid
  INTO TEMPORARY CourseIds
  FROM enrolled;
```

MySQL

```
CREATE TABLE CourseIds (
  SELECT DISTINCT cid FROM enrolled);
```

OUTPUT REDIRECTION

Insert tuples from query into another table:

→ Inner **SELECT** must generate the same columns as the target table.

→ DBMSs have different options/syntax on what to do with integrity violations (e.g., invalid duplicates).

SQL-92

```
INSERT INTO CourseIds
(SELECT DISTINCT cid FROM enrolled);
```

OUTPUT CONTROL

ORDER BY <column*> [ASC|DESC]

→ Order the output tuples by the values in one or more of their columns.

```
SELECT sid, grade FROM enrolled
 WHERE cid = '15-721'
 ORDER BY grade
```

sid	grade
53123	A
53334	A
53650	B
53666	D

OUTPUT CONTROL

ORDER BY <column*> [ASC|DESC]

→ Order the output tuples by the values in one or more of their columns.

```
SELECT sid, grade FROM enrolled
 WH
 OR
```

```
SELECT sid, grade FROM enrolled
 WHERE cid = '15-721'
 ORDER BY 2
```

OUTPUT CONTROL

ORDER BY <column*> [ASC|DESC]

→ Order the output tuples by the values in one or more of their columns.

```
SELECT sid, grade FROM enrolled
 WHERE cid = '15-721'
 ORDER BY 2
```

```
SELECT sid FROM enrolled
 WHERE cid = '15-721'
 ORDER BY grade DESC, sid ASC
```

sid
53666
53650
53123
53334

OUTPUT CONTROL

FETCH {FIRST|NEXT} <count> ROWS
OFFSET <count> ROWS

→ Limit the # of tuples returned in output.
→ Can set an offset to return a "range"

```
SELECT sid, name FROM student
 WHERE login LIKE '%@cs'
FETCH FIRST 10 ROWS ONLY;
```

OUTPUT CONTROL

FETCH {FIRST|NEXT} <count> ROWS
OFFSET <count> ROWS

→ Limit the # of tuples returned in output.
→ Can set an offset to return a "range"

```
SELECT sid, name FROM student
 WHERE login LIKE '%@cs'
FETCH FIRST 10 ROWS ONLY;
```

```
SELECT sid, name FROM student
 WHERE login LIKE '%@cs'
 ORDER BY gpa
OFFSET 10 ROWS
FETCH FIRST 10 ROWS WITH TIES;
```

WINDOW FUNCTIONS

Performs a calculation across a set of tuples that are related to the current tuple, without collapsing them into a single output tuple, to support running totals, ranks, and moving averages.

→ Like an aggregation but tuples are not grouped into a single output tuples.

How to "slice" up data
Can also sort tuples

```
SELECT FUNC-NAME(...) OVER (...)
  FROM tableName
```

Aggregation Functions
Special Functions

WINDOW FUNCTIONS

Aggregation functions:
→ Anything that we discussed earlier

Special window functions:
→ **ROW_NUMBER()** → # of the current row
→ **RANK()** → Order position of the current row.

sid	cid	grade	row_num
53666	15-445	C	1
53688	15-721	A	2
53688	15-826	B	3
53655	15-445	B	4
53666	15-721	C	5

```
SELECT *, ROW_NUMBER() OVER () AS row_num
  FROM enrolled
```

WINDOW FUNCTIONS

The **OVER** keyword specifies how to group together tuples when computing the window function.

Use **PARTITION BY** to specify group.

cid	sid	row_number
15-445	53666	1
15-445	53655	2
15-721	53688	1
15-721	53666	2
15-826	53688	1

```
SELECT cid, sid,
    ROW_NUMBER() OVER (PARTITION BY cid)
  FROM enrolled
 ORDER BY cid
```

WINDOW FUNCTIONS

You can also include an **ORDER BY** in the window grouping to sort entries in each group.

```
SELECT *,
    ROW_NUMBER() OVER (ORDER BY cid)
  FROM enrolled
 ORDER BY cid
```

WINDOW FUNCTIONS

Find the student with the second highest grade for each course.

Group tuples by cid
Then sort by grade

```
SELECT * FROM (
  SELECT *, RANK() OVER (PARTITION BY cid
            ORDER BY grade ASC) AS rank
    FROM enrolled) AS ranking
 WHERE ranking.rank = 2
```

NESTED QUERIES

Invoke a query inside of another query to compose more complex computations.

→ Inner queries can appear (almost) anywhere in query.

Outer Query

```
SELECT name FROM student WHERE
 sid IN (SELECT sid FROM enrolled)
```

Inner Query

NESTED QUERIES

Invoke a query inside of another query to compose more complex computations.

→ Inner queries can appear (almost) anywhere in query.

Outer Query

```
SELECT name FROM student WHERE
 sid IN (SELECT sid FROM enrolled)
```

Inner Query

```
SELECT sid,
       (SELECT name FROM student AS s
         WHERE s.sid = e.sid) AS name
  FROM enrolled AS e;
```

```
SELECT * FROM student
 ORDER BY (SELECT MAX(sid) FROM student);
```

NESTED QUERIES

Get the names of students in '15-445'

```
SELECT name FROM student
 WHERE ...
```

sid in the set of people that take 15-445

NESTED QUERIES

Get the names of students in '15-445'

```
SELECT name FROM student
 WHERE sid IN (
   SELECT sid FROM enrolled
    WHERE cid = '15-445'
 )
```

NESTED QUERIES

ALL→ Must satisfy expression for all rows in the sub-query.

ANY→ Must satisfy expression for at least one row in the sub-query.

IN→ Equivalent to '**=ANY()**' .

EXISTS→ At least one row is returned without comparing it to an attribute in outer query.

NESTED QUERIES

Find student record with the highest id that is enrolled in at least one course.

```
SELECT MAX(e.sid), s.name
  FROM enrolled AS e, student AS s
 WHERE e.sid = s.sid;
```

This won't work in SQL-92. It runs in SQLite, but not Postgres or MySQL (v8 with strict mode).

NESTED QUERIES

Find student record with the highest id that is enrolled in at least one course.

```
SELECT sid, name FROM student
 WHERE ...
```

"Is the highest enrolled sid"

NESTED QUERIES

Find student record with the highest id that is enrolled in at least one course.

```
SELECT sid, name FROM student
 WHERE sid is the
   SELECT MAX(sid) FROM enrolled
```

sid	name
53688	Bieber

NESTED QUERIES

Find all courses that have no students enrolled in it.

```
SELECT * FROM course
 WHERE ...
```

"with no tuples in the enrolled table"

cid	name
15-445	Database Systems
15-721	Advanced Database Systems
15-826	Data Mining
15-799	Special Topics in Databases

sid	cid	grade
53666	15-445	C
53688	15-721	A
53688	15-826	B
53655	15-445	B
53666	15-721	C

NESTED QUERIES

Find all courses that have no students enrolled in it.

```
SELECT * FROM course
 WHERE NOT EXISTS(
      tuples in the enrolled table
)
```

NESTED QUERIES

Find all courses that have no students enrolled in it.

```
SELECT * FROM course
 WHERE NOT EXISTS(
   SELECT * FROM enrolled
    WHERE course.cid = enrolled.cid
)
```

cid	name
15-799	Special Topics in Databases

LATERAL JOINS

The **LATERAL** operator allows a nested query to reference attributes in other nested queries that precede it.

→ You can think of it like a **for** loop that allows you to invoke another query for each tuple in a table.

```
SELECT * FROM
  (SELECT 1 AS x) AS t1,
  LATERAL (SELECT t1.x+1 AS y) AS t2;
```

t1.x	t2.y
1	2

LATERAL JOIN

Calculate the number of students enrolled in each course and the average GPA. Sort by enrollment count in descending order.

```
SELECT * FROM course AS c,
  For each course:
    ➡ Compute the # of enrolled students
  For each course:
    ➡ Compute the average gpa of enrolled students
```

LATERAL JOIN

Calculate the number of students enrolled in each course and the average GPA. Sort by enrollment count in descending order.

```
SELECT * FROM course AS c,
  LATERAL (SELECT COUNT(*) AS cnt FROM enrolled
           WHERE enrolled.cid = c.cid) AS t1,
  LATERAL (SELECT AVG(gpa) AS avg FROM student AS s
            JOIN enrolled AS e ON s.sid = e.sid
           WHERE e.cid = c.cid) AS t2;
```

LATERAL JOIN

Calculate the number of students enrolled in each course and the average GPA. Sort by enrollment count in descending order.

```
SELECT * FROM course AS c,
  LATERAL (SELECT COUNT(*) A
           WHERE enrolled.cid = c.cid) AS t1,
  LATERAL (SELECT AVG(gpa) AS avg FROM student AS s
            JOIN enrolled AS e ON s.sid = e.sid
           WHERE e.cid = c.cid) AS t2;
```

cid	name	cnt	avg
15-445	Database Systems	2	3.75
15-721	Advanced Database Systems	2	3.95
15-826	Data Mining	1	3.9
15-799	Special Topics in Databases	0	null

COMMON TABLE EXPRESSIONS

Specify a temporary result set that can then be referenced by another part of that query.

→ Think of it like a temp table just for one query.

Alternative to nested queries, views, and explicit temp tables.

```
WITH cteName AS (
   SELECT 1
)
SELECT * FROM cteName
```

COMMON TABLE EXPRESSIONS

You can bind/alias output columns to names before the **AS** keyword.

```
WITH cteName (col1, col2) AS (
   SELECT 1, 2
)
SELECT col1 + col2 FROM cteName
```

Postgres

```
WITH cteName (colXXX, colXXX) AS (
   SELECT 1, 2
)
SELECT * FROM cteName
```

COMMON TABLE EXPRESSIONS

Find student record with the highest id that is enrolled in at least one course.

```
WITH cteSource (maxId) AS (
   SELECT MAX(sid) FROM enrolled
)
SELECT name FROM student, cteSource
 WHERE student.sid = cteSource.maxId
```

OTHER THINGS TO NOTE

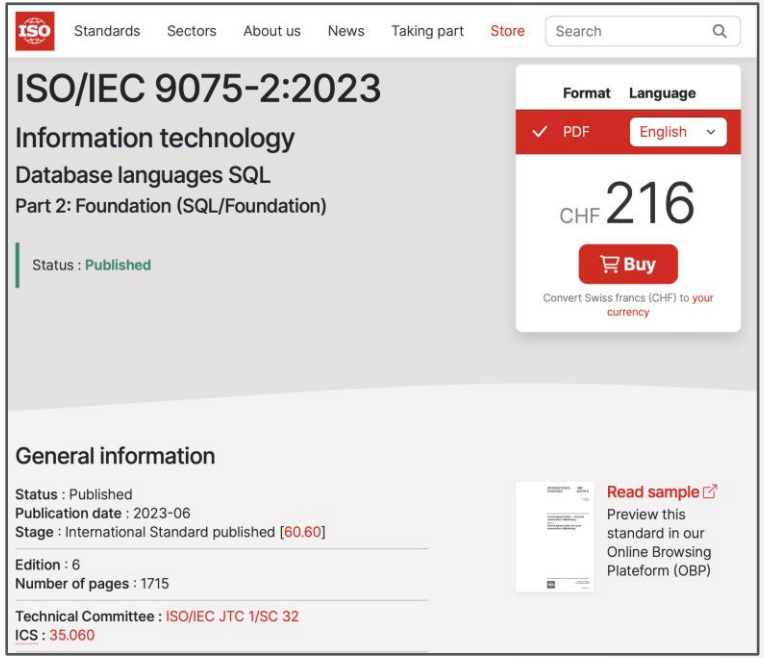
Identifiers (e.g. table and column names) are case-insensitive.

→ Makes it harder for applications that care about case (e.g., use CamelCased names).

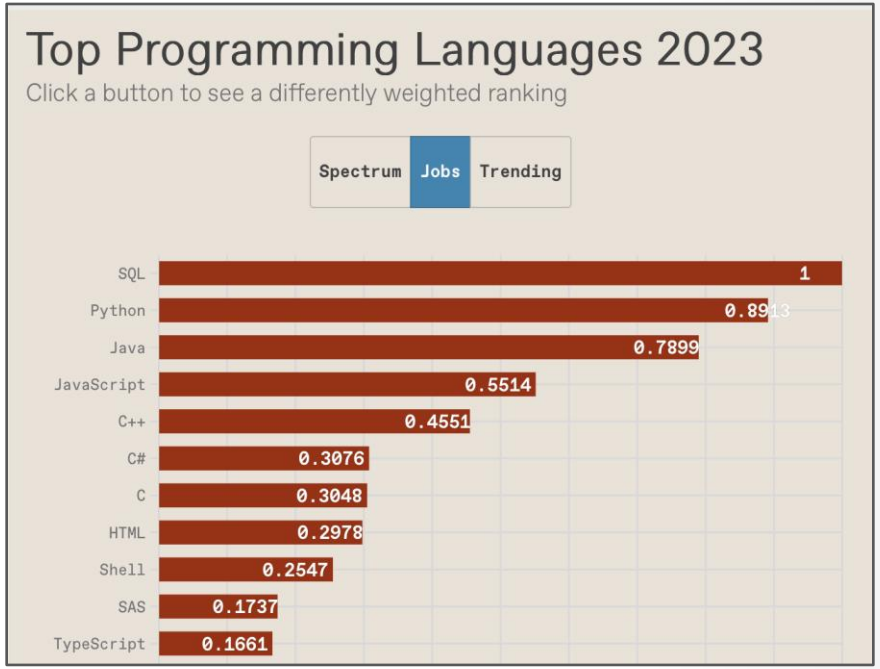
One often sees quotes around names:

→ `SELECT "ArtistList.firstName"`

You have to pay cash money to get the standard documents.



CONCLUSION

SQL is a hot language.
→ Lots of NL2SQL tools, but writing SQL is not going away.

You should (almost) always strive to compute your answer as a single SQL statement.